Granular Instrumental Variables

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Endogenous aggregate shocks confound the effect of large firms

The Goal: Estimate the causal elasticity of wages to labor demand.

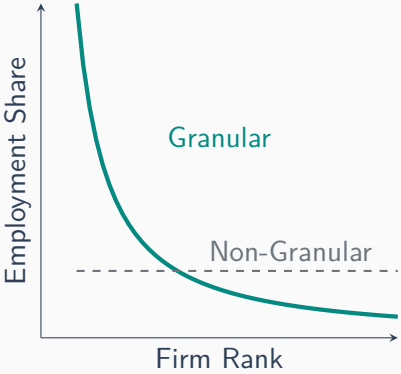
The Identification Challenge: **Simultaneity**

- **Supply Equation:** Wages respond to labor demand and macro supply shocks.
- **Demand Equation:** Labor demand responds to wages and macro demand shocks.

The Law of Large Numbers fails in modern economies

1. In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ ▸ Proof

2. In a **granular** economy, where the size distribution of firms follows a power law, a few firms are so large that their idiosyncratic luck drives aggregate volatility.



A Simple Model

A simple model of aggregate and idiosyncratic shocks

Consider an economy with N firms.

The system is given by:

$$w_t = \psi L_{St} + \varepsilon_t \tag{Supply}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \tag{Demand}$$

Where:

- w_t is the wage
- $L_{St} = \sum_i S_i L_{it}$ is the size-weighted average number of workers
- η_t and ε_t are aggregate shocks
- ψ is the labor supply elasticity, ϕ^d is the labor demand elasticity
- u_{it} is the idiosyncratic shock to firm i

We cannot estimate ψ and ϕ^d by OLS (simultaneity).

Constructing the Granular Instrumental Variable (z_t)

The GIV exploits the difference between size-weighted and equal-weighted averages.

Define the equal-weighted average outcome:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it} \qquad \text{(Equal-Weighted Average)}$$

The GIV is the size-weighted outcome minus the equal-weighted outcome:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \qquad \text{(GIV)}$$

The GIV isolates purely idiosyncratic firm shocks

Substitute the demand equation into the aggregate definitions: Proof

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$
$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

The common components $(\phi^d w_t + \eta_t)$ difference out:

$$z_t = u_{\Gamma t} = u_{St} - u_{Et} = \underbrace{\sum S_i L_{it}}_{\text{Size-Weighted}} - \underbrace{\frac{1}{N}\sum L_{it}}_{\text{Equal-Weighted}} \tag{GIV}$$

z_t is made *only* of idiosyncratic shocks.

GIV estimates elasticities via 2SLS

- **First Stage:** Regress aggregate labor demand (L_{St}) on the granular instrument (z_t). This extracts $\widehat{L}_{St}$, the variation driven *only* by the idiosyncratic shocks of firms.
- **Second Stage:** Regress aggregate wages (w_t) on the predicted labor demand from the First Stage ($\widehat{L}_{St}$) to obtain an unbiased estimate of ψ.

Validating the Instrument: Relevance and Exclusion

For z_t to be a valid instrument, two fundamental assumptions must hold in a granular economy.

- **Relevance (Cov$(L_{St}, z_t) \neq 0$):**

  The granular shocks must actually move the aggregate market. This requires a *concentrated market* (e.g., high Herfindahl index) where the idiosyncratic shocks of the largest firms are volatile enough to shift total labor demand.

- **Exclusion Restriction (Cov$(z_t, \varepsilon_t) = 0$):**

  The idiosyncratic shocks of large firms (z_t) must not directly affect aggregate wages (w_t) through any channel other than aggregate labor demand (L_{St}). They must be truly orthogonal to macro aggregate supply shocks (ε_t).

The GIV yields both supply and demand elasticities

By exogeneity, $\mathbb{E}[u_t \varepsilon_t] = 0 \implies \mathbb{E}[z_t \varepsilon_t] = 0$, where $u_t = (u_{it})_{i=1,\ldots,N}$.

Using the supply equation, $\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$: Proof

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \qquad \text{(Supply Elasticity)}$$

Using the equal-weighted demand equation, $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$:

$$\phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]} \qquad \text{(Demand Elasticity)}$$

The Full Model

Handling heterogeneous factor loadings

The Full Factor Model for granular firms

We now allow for heterogeneous loadings λ_i:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$$

The simple GIV $(L_{St} - L_{Et})$ assumes every firm has the same loading $\lambda_i = 1$. If large firms are more sensitive to the Fed, the simple GIV is **"leaky"**—it correlates with η_t. The simple GIV becomes: Proof

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** $(\mathbb{E}[z_t \eta_t] \neq 0)$.

Why not just Firm Size (S_i)?

- **Power vs. Validity:** Firm size (S_i) gives the instrument its **power**. We want the instrument to be large, and only large firms can create aggregate "granular" ripples.
- **The Conflict:** If we use $z_t = \sum(S_i - 1/N)L_{it}$, we have power, but we lack validity if S_i is correlated with λ_i (e.g., if big firms are more sensitive to macro shocks).
- **The "Tilt":** We start with firm sizes S_i but then "tilt" or project them until they are perpendicular (orthogonal) to the macro-sensitivities λ_i.
- **The math:** In matrix form, we want Γ to be as close as possible to S but satisfy $\Gamma \perp \lambda$. This is a standard projection problem: $\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S$
- **Intuition:** We take the firm sizes and subtract the portion of size that is explained by macro-sensitivity. What remains is "Pure Granular Weight."

The Solution: Orthogonal Optimal Weights Γ

Substitute Equation $L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$ into the definition of the weighted GIV:

$$\begin{aligned} z_t &= \sum \Gamma_i (\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t (\sum \Gamma_i) + \eta_t (\sum \Gamma_i \lambda_i) + \sum \Gamma_i u_{it} \end{aligned}$$

To ensure z_t contains *only* idiosyncratic shocks (u_{it}), we must satisfy two conditions:

1. $\sum \Gamma_i = 0$: This removes the endogenous wage w_t.
2. $\sum \Gamma_i \lambda_i = 0$: This removes the common macro shock η_t.

We need weights Γ such that the factor components **cancel exactly**. This ensures $z_t = \sum \Gamma_i u_{it}$ is purely idiosyncratic. For this, we use the $N \times N$ projection matrix Q that is orthogonal to the factor loadings—that is, $Q\lambda = 0$:

$$Q = I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

The optimal weights are then given by $\Gamma^{*\prime} = S'Q$. ▸ Propositions

Application

GIV in the Israeli Labor Market

The Israeli Tech Sector is highly granular

The Israeli high-tech industry is dominated by a few massive multinational R&D centers.

- **High Concentration:** Companies like Intel, Google, Microsoft, and NVIDIA account for a substantial, disproportionate share of tech employment.
- **A Heavy Tail:** The size distribution of tech employers in Israel exhibits a strong power law, yielding a high **excess Herfindahl index (h)**.
- **Macro vs. Micro:** It is difficult to separate general Israeli macroeconomic conditions (e.g., security shocks, interest rates) from the hiring dynamics of these specific giants.

Applying GIV to Israeli tech wages

How do we isolate the causal spillover of these tech giants on aggregate Israeli wages?

- **The Idiosyncratic Shocks (u_{it}):** We exploit global shocks specific to the multinationals that are arguably exogenous to the Israeli macroeconomy (e.g., global restructuring at Intel, a worldwide AI hiring boom at NVIDIA).
- **The Instrument (z_t):** Construct the GIV by taking the size-weighted employment shocks of these giants and projecting out the common Israeli macroeconomic factor loadings.
- **The 2SLS Estimation:** Regress aggregate Israeli wages on the component of labor demand driven *only* by these isolated, granular tech shocks.

Takeaways: GIV unconfounds local macro dynamics

The GIV method solves the simultaneity problem in concentrated labor markets.

- **Causal Wage Spillovers:** We can measure exactly how much a hiring freeze at a major tech firm depresses aggregate local wages, separated from a general economic downturn.
- **Beyond the US:** The method is perfectly suited for small, open economies like Israel, where granular actors often drive aggregate volatility.

Appendix A

Derivations

Aggregate volatility decays at $1/\sqrt{N}$ under equal weights

Suppose there are N firms, each receiving an independent idiosyncratic shock u_{it}. Assume these shocks have mean 0 and a common variance σ^2:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$

Deriving the Size-Weighted Aggregate (L_{St})

The size-weighted outcome is defined as $L_{St} = \sum_{i=1}^{N} S_i L_{it}$, where S_i are weights such that $\sum S_i = 1$.

$$L_{St} = \sum_{i=1}^{N} S_i(\phi^d w_t + \eta_t + u_{it})$$

$$L_{St} = \sum_{i=1}^{N} S_i \phi^d w_t + \sum_{i=1}^{N} S_i \eta_t + \sum_{i=1}^{N} S_i u_{it}$$

Since ϕ^d, w_t, and η_t do not depend on the index i, we can pull them out:

$$L_{St} = \phi^d w_t \left(\sum_{i=1}^{N} S_i \right) + \eta_t \left(\sum_{i=1}^{N} S_i \right) + \sum_{i=1}^{N} S_i u_{it}$$

Using the property $\sum S_i = 1$ and defining the size-weighted idiosyncratic shock as $u_{St} = \sum S_i u_{it}$, we get:

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$

Deriving the Equal-Weighted Aggregate (L_{Et})

The equal-weighted outcome is defined as $L_{Et} = \frac{1}{N}\sum_{i=1}^{N} L_{it}$.

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}(\phi^d w_t + \eta_t + u_{it})$$

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}\phi^d w_t + \frac{1}{N}\sum_{i=1}^{N}\eta_t + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Again, pulling constants out of the summations:

$$L_{Et} = \phi^d w_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \eta_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Since $\frac{1}{N}\sum_{i=1}^{N}1 = \frac{1}{N}\cdot N = 1$, and defining the equal-weighted idiosyncratic shock as $u_{Et} = \frac{1}{N}\sum u_{it}$, we get:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

Identifying Supply Elasticity (ψ)

To find ψ, we use z_t as an instrument for L_{St} in the supply equation. We take the expectation of the product of the supply equation and the instrument:

$$\mathbb{E}[w_t z_t] = \mathbb{E}[(\psi L_{St} + \epsilon_t) z_t]$$
$$\mathbb{E}[w_t z_t] = \psi \mathbb{E}[L_{St} z_t] + \mathbb{E}[\epsilon_t z_t]$$

By the exogeneity condition, $\mathbb{E}[\epsilon_t z_t] = 0$. Thus:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \tag{1}$$

This is the standard IV formula where z_t instruments for L_{St}.

Identifying Demand Elasticity (ϕ^d)

To find ϕ^d, we use the equal-weighted aggregate demand equation derived on slide 8:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et} \tag{41}$$

We aim to show that $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$. Substituting the demand equation:

$$\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = \mathbb{E}[(\eta_t + u_{Et}) z_t] = \mathbb{E}[\eta_t z_t] + \mathbb{E}[u_{Et} z_t] \tag{2}$$

We already know $\mathbb{E}[\eta_t z_t] = 0$. Now we examine $\mathbb{E}[u_{Et} z_t]$. Under the assumption that u_{it} are i.i.d. with variance σ_u^2:

$$\begin{aligned} \mathbb{E}[u_{Et} z_t] &= \mathbb{E}[u_{Et}(u_{St} - u_{Et})] \\ &= \mathbb{E}[u_{Et} u_{St}] - \mathbb{E}[u_{Et}^2] \end{aligned}$$

Identifying Demand Elasticity (ϕ^d)

$$\mathbb{E}[u_{Et}u_{St}] - \mathbb{E}[u_{Et}^2] =$$

$$= \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)\left(\sum_j S_j u_{jt}\right)\right] - \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)^2\right]$$

$$= \frac{1}{N}\sum_i S_i\sigma_u^2 - \frac{1}{N^2}\sum_i \sigma_u^2$$

$$= \frac{1}{N}\sigma_u^2(1) - \frac{1}{N}\sigma_u^2 = 0$$

Since $\mathbb{E}[u_{Et}z_t] = 0$, it follows that $\mathbb{E}[(L_{Et} - \phi^d w_t)z_t] = 0$. Rearranging:

$$\mathbb{E}[L_{Et}z_t] = \phi^d\mathbb{E}[w_t z_t]$$

$$\phi^d = \frac{\mathbb{E}[L_{Et}z_t]}{\mathbb{E}[w_t z_t]}$$

This identifies the demand elasticity using the same granular instrument z_t.

Derivation: Why the Simple GIV Fails

The simple GIV is defined as $z_t^{simple} = L_{St} - L_{Et}$. Let's derive its components:

1. **Size-weighted average (L_{St}):**

$$\begin{aligned} L_{St} &= \sum S_i L_{it} = \sum S_i(\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t \underbrace{\sum S_i}_{1} + \eta_t \underbrace{\sum S_i \lambda_i}_{\lambda_S} + \underbrace{\sum S_i u_{it}}_{u_{St}} \\ &= \phi w_t + \lambda_S \eta_t + u_{St} \end{aligned}$$

2. **Equal-weighted average (L_{Et}):**

$$\begin{aligned} L_{Et} &= \frac{1}{N} \sum L_{it} = \phi w_t + \eta_t \underbrace{\frac{1}{N} \sum \lambda_i}_{\lambda_E} + u_{Et} \\ &= \phi w_t + \lambda_E \eta_t + u_{Et} \end{aligned}$$

3. **The Resulting GIV:**

$$z_t^{simple} = L_{St} - L_{Et} = (\lambda_{\mathbf{S}} - \lambda_{\mathbf{E}})\eta_{\mathbf{t}} + (u_{St} - u_{Et})$$

Appendix B

Propositions

Granular Instrumental Variables isolate idiosyncratic shocks

The GIV z_t is constructed from observables, L_{it}, using weights $\Gamma \in \mathbb{R}^N$ orthogonal to factor loadings:

$$z_t := \Gamma' L_t = \sum_{i=1}^{N} \Gamma_i L_{it}$$

By construction, this isolates a linear combination of idiosyncratic shocks u_t:

$$z_t = \Gamma' u_t \quad \Rightarrow \quad \text{exogenous to aggregate shocks } \eta_t, \varepsilon_t$$

This allows us to construct the sample estimators:

$$\hat{\psi}_T = \frac{\sum_{t=1}^{T} w_t z_t}{\sum_{t=1}^{T} L_{St} z_t} \qquad \hat{\phi}_T^d = \frac{\sum_{t=1}^{T} L_{\tilde{E}t} z_t}{\sum_{t=1}^{T} w_t z_t}$$

Moment conditions link structural equations to the GIV

The estimators follow directly from the exogeneity of z_t to structural shocks:

Inverse Supply Elasticity

From $w_t = \psi L_{St} + \cdots + \varepsilon_t$:

$$\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$$

$$\Rightarrow \quad \psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

Demand Elasticity

For weights $\tilde{E}$ where $\mathbb{E}[u_{\tilde{E}t} u_{\Gamma t}] = 0$:

$$\mathbb{E}[(L_{\tilde{E}t} - \phi^d w_t) z_t] = 0$$

$$\Rightarrow \quad \phi^d = \frac{\mathbb{E}[L_{\tilde{E}t} z_t]}{\mathbb{E}[w_t z_t]}$$

Proposition 1: The GIV estimator is asymptotically consistent

Under standard regularity conditions, as the number of time periods $T \to \infty$:

Inverse Supply Elasticity

$$\hat{\psi}_T \xrightarrow{\text{a.s.}} \psi$$

Demand Elasticity

$$\hat{\phi}_T^d \xrightarrow{\text{a.s.}} \phi^d \quad (\text{if } \psi \neq 0)$$

Intuition: The orthogonal weighting Γ ensures the instrument is uncorrelated with the structural error term, resolving simultaneity bias.

Pass-through multipliers capture shock amplification

To understand the variance of these estimators, we define two key multipliers:

Quantity Pass-through

$$M := \frac{1}{1 - \psi\phi^d}$$

Price Pass-through

$$\mu := \psi M$$

These represent the pass-through from idiosyncratic and aggregate shocks to the aggregate quantity and the price, respectively.

Proposition 2: Asymptotic variance scales with aggregate noise

The estimators converge in distribution to a normal distribution:

$$\sqrt{T}(\hat{\psi}_T - \psi) \xrightarrow{d} \mathcal{N}(0, \sigma_\psi^2), \quad \sigma_\psi = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\psi}}{|M|}$$

$$\sqrt{T}(\hat{\phi}_T^d - \phi^d) \xrightarrow{d} \mathcal{N}(0, \sigma_{\phi^d}^2), \quad \sigma_{\phi^d} = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\phi}}{|\mu|}$$

High idiosyncratic volatility improves estimator precision

The variance equations reveal the signal-to-noise ratio inherent to GIV:

Signal Strength

$|\mathbb{E}[u_{St}u_{\Gamma t}]|$

Larger granular shocks increase relevance.

Instrument Noise

$\mathbb{E}[u_{\Gamma t}^2]^{1/2}$

Variance of the constructed GIV itself.

Structural Noise

$\sigma_\varepsilon/|M|$

Unobserved shocks dampening precision.

Proposition 3: Optimal weights use an orthogonal projection

To minimize the asymptotic variances σ^2_ψ and $\sigma^2_{\phi^d}$, we use the $N \times N$ projection matrix Q that is orthogonal to the factor loadings λ:

$$Q := I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

The optimal weights vector Γ^* is the projection of the size vector S:

$$\Gamma^* = S'Q$$

Result: Γ^* is the vector closest to S, while remaining orthogonal to the unobserved aggregate factor loadings.